

Improving Ergonomics and Comfort of Napping in EngSci Common Room Chairs

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ESC101 - Praxis I

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Introduction

This design brief will explore the current inadequacies faced when napping on the University of Toronto's Engineering Science (EngSci) common room chairs; it will then outline objectives for a more ergonomic, comfortable and portable solution that can be integrated with the chairs. Currently, students complain of back and neck pain after napping in these chairs [1], indicating a lack of comfort and proper ergonomics for taking a nap. Over time, this can result in lasting detriments to the health of students [2]. Thus, access to an ergonomic napping solution in the common room is vital to improve the energy level of sleep deprived EngSci students without causing long-term health detriments.

Background

1. Napping Definition

- a. Napping will be defined as a light and brief sleep that lasts less than 30 minutes [1][3]. Body positions for napping can be more varied than those for sleeping because of the shorter time frame, so designs are not required to act as a traditional mattress or offer entirely horizontal support for the body.

2. Environment Definition and Reasoning

- a. The working environment of the design will be the chairs in the EngSci common room, not including high chairs or couches. Figure 1 shows an example chair in the EngSci common room for the design. Designs should not seek to replace the chair, but integrate with it.
- b. The environment has been limited to the EngSci common room chairs because entry requires an EngSci student's T-Card. This means that safety during the nap is less of a concern for the design.
- c. Since it is not socially acceptable to sleep on the floor, lie down across a couch or on a table to sleep; designs should not rely on these surfaces.



Figure 1 - Chair from the Engineering Science Common Room. Designs may use the chair to support the user's body weight. See [A.1] for details.

Stakeholder Expectations

This section will discuss all relevant stakeholders for the design of a napping device, and list what they want in a design.

1. UofT Engineering Science Students

EngSci students are the primary stakeholders because they are the ones napping in the common room [1]. They want a design that can integrate with the existing chairs to make them more ergonomic and comfortable, without replacing the original chairs. Students also want a solution that is portable enough to be carried around campus, because they must travel between classes and home. Finally, they want a design that requires minimal cleaning effort to stay hygienic.

2. People within the vicinity of the product

These secondary stakeholders want designs to take up minimal space in the common room.

3. Design Team

Need, Goals, and Objectives

Primary Stakeholder Need

UofT Engineering Science students need an attachment to Engineering Science Common Room chairs to make napping more comfortable and ergonomic, while also being portable and hygienic.

To satisfy this need, four goals with their associated objectives will be identified and justified.

1. **Provides an ergonomic resting place**

The current common room chairs are not ideal for napping due to their structure having backrests of angles 80°-90°. Thus, the design will allow EngSci students to sleep with their backs/necks straight instead of hunched over, preventing chronic back and neck problems in the long term.

	Description	Justification
Objective 1.1	Back angle of 20°-60° from the vertical	An angle of 60° produced the highest quality sleep and anything greater than 20° was an improvement [5].
Objective 1.2	Neck angle does not drop more than 10° relative to spine	It is important to keep the head upright and support it, especially when sleeping upright like in a car or plane to prevent chronic neck pain [6]. Having the neck at an angle greater than 10° for

		prolonged periods of time was found to be a potential source of neck pain among other chronic issues [7].
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2. Design for comfortability

Since one main desire of the primary stakeholders is a comfortable place to sleep, designs should be made to prioritize this. Comfort is described by the Oxford Dictionary, as ‘freedom from pain’ [8]. Therefore, designs should aim to reduce any pain caused by high pressure or hard resting surfaces. The user should also be relaxed, meaning that no voluntarily controlled skeletal muscles need to contract for the user’s body to remain still. The following objectives address those ways to increase comfort.

	Description	Justification
Objective 2.1	Gauge pressure from the support surface must not exceed 31.8 kPa at any part of the skin.	The user will not feel pain below this threshold, because 31.8 kPa applied to the back or legs is the threshold to make a user on a chair uncomfortable [9]. The pressure threshold for pain on other parts of the body is larger [10], so a minimum of 31.8 kPa is enough to ensure comfort.
Objective 2.2	Material that contacts and supports the body must be rated less than 45 ILD.	45 ILD surfaces will provide a comfortable surface for napping because 44 ILD is the hardest firmness available for sleeping surfaces like mattresses [11].
Objective 2.3	All of the user's skeletal muscles [A.2] register less than 3 μ V above signal baseline on an EMG (EMG amplitude $\leq$ baseline + 3 μ V) when resting with design.	Muscle responsiveness can be measured by electromyography (EMG). A threshold of 3 μ V above signal baseline on EMGs was found to be the best indicator that muscles were active [12], and all voluntary skeletal muscles should be at rest, so they should be below this threshold.

3. The design should be portable

EngSci students walk around campus to attend their classes. As the location of lecture halls are spread among various buildings that require 5-10 minutes of walking, the design should be portable for students to carry to and around campus without unnecessary

physical strain. Thus, allowing sleep-deprived engineering students to easily use the design to achieve high-quality sleep.

	Description	Justification
Objective 3.1	Mass under 4.4 kg.	The maximum portable weight for one-hand carry of the smallest 5th-percentile of population is 4.4 kg [13]. Heavier weights require support from waist/harness rather than free carry. Designing under that guideline allows for comfort, easy grip, and free of physical strains.
Objective 3.2	Under one-hand carrying volume: 675 cm ³ .	The maximum volume of portable devices to be carried one-handed is 675 cm ³ [13]. Larger load volume/shape contributes to user discomfort.
Objective 3.3	Under standard backpack internal dimensions at smallest size: 43cm×25cm×15cm.	The dimensions of 43cm×25cm×15cm correspond to the standard school backpack internal dimensions recommended by studies on college students [15]. Designing under that dimension criteria ensures a comfortable fit in a student backpack.

4. The design stays hygienic with minimal cleaning effort

In order for EngSci students to feel comfortable while taking a nap, the product has to be hygienic; as dirt causes psychological discomfort [16]. By making the product antibacterial and stain resistant (oleophobic and hydrophobic) [17], EngSci students can spend minimal time cleaning the product.

	Description	Justification	Metric
Objective 4.1	All plastic and non porous surfaces should have an antibacterial activity value (R) greater than or equal to	Many people commute to the UofT St George campus, increasing the diversity and abundance of bacteria on campus. An R value of 2.0	R

	2.0 [18] per ISO 22196 [19].	indicates a 99% reduction of bacteria in 24 hours on porous surfaces, allowing EngSci's to worry less about sanitizing the product.	
Objective 4.2	All fabric should have an antibacterial value (A) greater than or equal to 2; falling under the "significant" category of efficacy of antibacterial property in ISO 20743 [20].	Germ on the product can easily enter the user's body through skin contact or particle inhalation. With an antibacterial value of 2.0 or greater, definite antibacterial effects can be confirmed [20], inhibiting significant growth of bacteria.	A
Objective 4.3	All fabric should have a result of no more than 1.0 grams when put through AATCC test method 35 [21].	Stains are time consuming to remove and harbour germs. With an AATCC 35 result of less than 1.0 grams, the tested fabric is hydrophobic.	Grams
Objective 4.4	All fabric should receive a grade of "A" per ISO 14419 [22].	Oleophobic fabric will prevent potential oil stains from food EngSci students eat near the product, as well as oils the human body naturally produces.	Assigned grade based on visually looking at the results.

Reference Designs

For each reference design, features that satisfy objectives will be discussed, as well as objectives not met. From this analysis the reader should gain an idea of what features may be useful to implement in a solution.

Portable Head Rest

One existing design is a portable head rest as shown in Figure 2. It consists of a supporting board that is inserted into a traveller's bag, with a hinge attached to the cushion.

It satisfies Objective 3.1 - 3.3 because it is within the portable dimensions ($21.59\text{cm} \times 15.24\text{cm} \times 2.22\text{cm} < 43\text{cm} \times 25\text{cm} \times 15\text{cm}$) [24]. The design takes advantage of features that the environment has, such as the chair and the user's bag, to support the user's body weight, so that it doesn't have to waste space on providing this support.

However, this design does not satisfy Goal 1-2 and Goal 4.



Figure 2 - Portable head support intended to be supported by the user's bag. Allows the user to sleep in a face-down position on a cushion [24].

Office Chair Support Cushion



Figure 3: Ergonomic head, neck, back, and leg rest secured on office chair. Designed to support users' spine and neck for comfortable sleep. [23]

An additional design to achieve comfortable rest on a typical office chair is shown in Figure 3. This design satisfies Goals 1-2 as it is made of memory foam cushioning with ideal pressures for blood circulation and firm support for the user's neck and back to be held upright (at ideal angles), with muscles relaxed. This design does not satisfy goal 4 but instead features slidable and detachable sheets (with hidden zipper) that allows for daily machine wash with strong detergents.

This design is a perfect example of the ergonomic features our goals aim for but has overlooked portability, therefore failing to meet the dimension criterias of Objectives 3.1-3.3.

Conclusion

Considering that unergonomic naps are so common in the EngSci common room, hopefully solutions inspired by this design brief can improve the ergonomics and comfort of those naps, while providing a portable and hygienic product at the same time. These goals and objectives were created after an analysis of EngSci student expectations to ensure that primary stakeholder wants were represented as best as possible. Therefore, any design under the outlined brief should transform ordinary office chairs in the EngSci common room into ergonomic napping environments, improving student rest quality and productivity.

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Source Extracts

Source Extract: How posture causes back pain [2]

Conclusion: This review emphasises the role of sleep posture in managing low back pain (LBP) by recommending supine and supportive side-lying positions for better spinal health. Prone sleeping, linked to pain, should be avoided. Ergonomic interventions, such as proper bedding and posture education, are suggested for LBP relief.

Taken from a study that analyzed sleeping positions and back pain. Used to determine if napping posture could lead to health detriments.

Source Extract: Back angle on a chair for optimal napping [5]

Highlights

- Self-reported sleep quantity and quality were similar between the reclined (60° recline) and flat seat (87° recline).
- Sleepiness increased in the reclined seat (60° recline), while sleepiness decreased in the flat seat (87° recline).
- Wake after sleep onset (WASO) was higher in the reclined seat (60° recline).
- Sleep latency was higher in the flat seat (87° recline).

This source extracts shows that an angle of 60° is the most optimal angle for sleeping on a reclined chair.

Source Extract: Effect of neck position and spine on neck pain [6]

As with so many things, when it comes to neck pain, an ounce of prevention may be worth a pound of cure. It's true that some causes of **neck pain**, such as age-related wear and tear, are not under your control. On the other hand, there are many things you can do to minimize your risk. One place to start is to look at how you sleep and what effect this may have on neck pain.

What is the best sleeping position for neck pain?

Two sleeping positions are easiest on the neck: on your side or on your back. If you sleep on your back, choose a rounded **pillow** to support the natural curve of your neck, with a flatter pillow cushioning your head. This can be achieved by tucking a small neck roll into the pillowcase of a flatter, softer pillow, or by using a special pillow that has a built-in neck support with an indentation for the head to rest in. Here are some additional tips for side- and back-sleepers:

1. Try using a feather pillow, which easily conforms to the shape of the neck. Feather pillows will collapse over time, however, and should be replaced every year or so.
2. Another option is a traditionally shaped pillow with "memory foam" that conforms to the contour of your head and neck. Some cervical pillows are also made with memory foam. Manufacturers of memory-foam pillows claim they help foster proper spinal alignment.
3. Avoid using too high or stiff a pillow, which keeps the neck flexed overnight and can result in morning pain and stiffness.
4. If you sleep on your side, keep your spine straight by using a pillow that is higher under your neck than your head.
5. When you are riding in a plane, train, or car, or even just reclining to watch TV, a horseshoe-shaped pillow can support your neck and prevent your head from dropping to one side if you doze. If the pillow is too large behind the neck, however, it will force your head forward.

This source extract shows that the neck dropping to any one side is a compromised position. Also explains that the neck needs to be supported, but if it is too large, it will force your head forward.

Source Extract: Neck angle of 10° or greater cause neck pain and other issues [7]

3 Pillows: Neck Support

When sleeping in the prone position, the head is usually turned to the side, which increases the rotation of the neck. This rotation is up to 90° and flexion of 10°–20° of the neck, with negative repercussions on muscular structures, nerves, vessels, ligaments, and other segments. This posture constitutes a potential source of neck pain, fatigue, muscle tension, tension headache, and pain in the temporomandibular joint [11]. Consequently, there is an increase in spinal load, because several facet joints of the most cranial vertebra are compressed on the ipsilateral side, while the contralateral ligaments will be under tension and stretching. Blood vessels can also be compressed, causing headaches, dizziness, and other disorders [1]. Despite all the disadvantages, many people prefer the prone position. It is therefore advisable to improve the alignment of the spine by placing a pillow, under the shoulder and the ribcage next to which the head is facing, or by raising your arm from that side, or even by placing a pillow under the belly to restore lumbar lordosis. The same alignment of an eventual hyperlordosis can be achieved by raising the knee and the hip next to which the head is rotated, or by placing a pillow under the hip and knee on this side. In fact, most of the aforementioned corrections implement a slight change of position to an almost lateral position [1].

This source extract explains that when lying on your side, any flexion of the neck greater than 10° showed negative repercussions on the body, such as neck pain. This is applicable to a person sitting because the neck is now upright and has

no support from anything on the side. Therefore, any potential harm caused by the neck being at a compromised angle while lying down is exacerbated when sitting upright with the neck at a compromised angle.

Source Extract: Force threshold for discomfort [9]

Table2. Average values (in N, not equalized) at which the 23 participants experience the pressure as no longer comfortable, with standard deviations (in brackets).

Backrest	10.92 (5.30)	12.17 (5.94)	15.17 (7.2)	12.61 (6.86)	10.73 (6.68)
		11.81 (8.3)	21.57 (16.02)	13.10 (7.99)	
		14.38 (9.44)	19.24 (13.1)	15.36 (9.73)	
		15.29 (6.59)	21.01 (8.67)	17.18 (8.67)	
Seat pan	22.39 (9.33)	23.42 (11.09)	23.21 (7.87)	22.69 (9.40)	
	23.5 (9.33)	22.39 (10.17)	21.35 (8.01)	22.64 (8.88)	
	19.76 (8.63)	15.73 (6.23)	16.61 (6.45)	18.83 (7.28)	

This table shows the average value of force applied to different parts of the user's body when they sit on a chair when they become uncomfortable. The force was measured by an advanced force gauge applied in a circle with a diameter of 20 mm. Data from the table was used to justify the pressure threshold the skin could experience before the user became uncomfortable.

Source Extract: Pressure threshold for pain[10]

Zone	Measured PDT (in kPa)				Normalised PDT (in percentage)			
	All mean (SD) (n=28)	Male mean (SD) (n=14)	Female mean (SD) (n=14)	Gender based difference p-value	All mean (SD) (n=28)	Male mean (SD) (n=14)	Female mean (SD) (n=14)	Gender based difference p-value
Scalp	190.7 (83.8)	167.3 (72.8)	214.2 (94.9)	p≤0.01**	74.8 (14.8)	75.6 (15.0)	74.1 (14.6)	p≤0.05*
Forehead	144.1 (69.4)	129.7 (62.9)	158.6 (75.9)	p≤0.01**	57.7 (18.7)	58.7 (18.1)	56.6 (19.4)	.42
Temple	151.2 (64.7)	125.2 (56.5)	177.3 (73)	p≤0.01**	59.6 (14.0)	57.7 (15.4)	61.5 (12.4)	.09
Back of head	196.1 (91.7)	165.4 (72.3)	226.8 (111.1)	p≤0.01**	74.5 (15.1)	72.2 (15.4)	77.0 (14.5)	p≤0.01**
Nose	67.1 (34.2)	59 (28.9)	75.3 (39.6)	p≤0.05*	27.5 (12.0)	27.8 (12.0)	27.3 (12.2)	.75
Cheekbone	92 (50.6)	80 (45.4)	104 (55.8)	p≤0.01**	35.9 (13.4)	35.8 (13.7)	36.0 (13.2)	.96
Cheek	80.1 (47.9)	66.5 (36.2)	93.8 (59.7)	p≤0.01**	30.5 (10.1)	29.6 (10.7)	31.4 (9.3)	.11
Mouth	67.3 (37.9)	53.6 (32.9)	81 (43)	p≤0.01**	26.7 (10.9)	24.5 (11.5)	28.8 (10.0)	p≤0.01**
Jaw-chin	70.3 (34.8)	61.1 (27.8)	79.5 (41.8)	p≤0.01**	28.8 (10.9)	30.2 (11.9)	27.5 (9.7)	.11
Neck front	65 (35.8)	56.1 (27.1)	73.9 (44.6)	p≤0.01**	26.4 (11.4)	26.9 (11.6)	25.8 (11.1)	.45
Neck back	121 (69.8)	108.7 (64.5)	133.3 (75.1)	p≤0.01**	47.0 (17.4)	48.3 (19.4)	45.7 (15.1)	.26

Data taken from a study that mapped the pressure discomfort threshold on the human head and neck. This table was also used to justify pressure threshold for objective 2.1

Source Extract: EMG measurements for muscles at rest[11]

Prolonged sedentary behaviour is detrimental to health due to low contractile activity in large lower extremity muscle groups. This muscle inactivity can be measured with electromyography (EMG), but it is unknown how methodological factors affect responsiveness longitudinally. This study ranks 16 different EMG inactivity thresholds based on their responsiveness (absolute and standardized effect size, responsiveness) using data from a randomized controlled trial targeted at reducing and breaking up sedentary time (InPact, ISRCTN28668090). EMG inactivity duration and usual EMG inactivity bout duration (weighted median of bout lengths) were measured from large lower extremity muscle groups (quadriceps, hamstring) with EMG-sensing shorts. **The results showed that the EMG inactivity threshold above signal baseline (3 μ V) provided overall the best responsiveness indices.** At baseline, EMG inactivity duration of $66.8 \pm 9.6\%$ was accumulated through 73.9 ± 36.0 s usual EMG inactivity bout duration, both of which were reduced following the intervention (-4.8 percentage points, -34.3 s). The proposed methodology can reduce variability in longitudinal designs and the detailed results can be used for sample size calculations. Reducing EMG inactivity duration and accumulating EMG inactivity in shorter bouts has a potential influence on muscle physiology and health.

This source was used to find the best EMG reading boundary between active and resting muscles. A boundary like this has not been standardized, but this source found that responsiveness indices for 3 μ V above EMG signal baseline performed the best.

Source Extract: Firmness of sleeping surfaces, measured in ILD [12]

ILD FIRMNESS GUIDE

- Plush has a range of 16.5 ILD to 22.5 ILD
- Soft has a range of 22.5 ILD to 26.5 ILD
- Medium has a range of 26.5 ILD to 30.5 ILD
- Firm has a range of 30.5 ILD to 34.5 ILD
- Extra-Firm has a range of 34.5 to 38.5 ILD

Most people sleep comfortably on mattresses that have a 25-35 ILD.

There is no standard defined for ILD ratings of mattresses, so there isn't a super reliable source for recommended ILD for sleeping surfaces. This website provided one possible perspective that was used to define objective 2.2, but this means that the objective could be biased.

Source Extract: One hand carry weight [13]

The 5th percentile weight (male and female) for a portable product (one hand carry) is 4.4 kg. (Note that for portable computers, the arbitrary division of 8.5 kg separates the lightweight notebooks from the heavier laptops [22]).

This source extract explains that the maximum acceptable portable weight for one-hand carry of the smallest 5th-percentile of the human population is approximately 4.4 kg.

Source Extract: Portable one-handed devices [13]

For portable devices to be carried one handed, the maximum dimensions are:

- Length 1000 mm.
- Width 150 mm.
- Depth 450 mm. (for ground clearance)

This source extract explains that the maximum volume of portable devices to be carried one handed is 675 cm³.

Source Extract: Backpack dimensions [15]

3. Experimental Protocol

3.1. Backpack Development

3.1.1. Backpack Height

The height of the backpack is acquired from the student's sitting shoulder height. It is recommended that the maximum height of the backpack should not exceed ten centimeters from the fifth percentile of the student's sitting shoulder height. Then, as recommended in Mououdi et al. [30], Larisang [47], and Kristina and Amanda [48], backpack height is equal to subtraction of 2.5% of the thigh thickness from the 5% of the sitting shoulder height. Then, backpack height is equal to 5% of the sitting shoulder height minus 2.5% of the thigh thickness. This value is equivalent to 43 cm (e.g., 53.3-10.1). Since there are three pockets, and the length of the textbooks is ranged between 21.5 cm and 27 cm. Therefore, the backpack height is 30 cm.

3.1.2. Backpack Width

The width of the backpack ought not to surpass the width of the student's body with the goal that the student can move all-around more freely. Thus, the width of the backpack is measured by determining the distance between the shoulders. The shoulder joint is simply the body width range. It is recommended that the maximum width of the backpack should not exceed the fifth percentile of the student's shoulder to shoulder distance. Then, as recommended in Mououdi et al. [30], Larisang [47], and Kristina and Amanda [48], the backpack width is equal to two-thirds of the 5% of the shoulder to shoulder length. Then, backpack width is equal to two-thirds of the 5% of the shoulder to shoulder length. This value is equivalent to 25 cm (e.g., 2/3 of the 37.3).

3.1.3. Backpack Thickness

First, determine the maximum load of the backpack, which is based on 10% of the students' weights. 10% of the average weight of college students was 75.5 kg, as shown in Table 1. The volume of the backpack is acquired from the transformation of the most allowable backpack weight that has been determined earlier into backpack volume. Second, convert that weight to volume based on the following: 1 kg of paper materials (standard) can be converted to 832.8 cm³ [48], so later can be resolved the thickness of the backpack that can just suit the load as per the most allowable backpack weight. This volume is then analyzed to obtain the thickness of the backpack utilizing the volume equation (2).

$$\begin{aligned} \text{Pocket volume} &= \text{Total required volume}/3 = \text{Height} \times \text{Width} \times \text{Thick} \\ \text{Pocket volume} &= \text{total volume}/3 = \text{Height} \times \text{Width} \times \text{Thick} \\ &= 10/100 * 75.5 * 832.8/3 = 30 * 25 * \text{Thick} \end{aligned} \quad (2)$$

Thickness ≈ 3 cm. 66% Allowance was added for materials that may has less specific gravity than paper. Then, thickness for each pocket = 5 cm.

This source extract explains that the dimensions of 43 cm × 25 cm × 15 cm correspond to the standard backpack maximum internal dimensions recommended by studies, where the maximum height, width, and thickness is determined based on research findings of the 5th percentile of the student's sitting shoulder height, the 5th percentile of the student's shoulder to shoulder distance, and 10% of the average weight of college students, respectively.

Source Extract: Dirt Causes Psychological Discomfort [16]

We will not understand hygiene if we do not also discuss disgust. Dirt can be disgusting; removing it is, therefore, not only a social act but also a psychological and emotional one. But what is dirt and why does it cause disgust? In a classic essay, the anthropologist, Mary Douglas, defined dirt as "matter out of place."⁹ Her central thesis was that "absolute" dirt does not exist. The same thing or act can be clean as well as dirty. It all depends on the context. Red wine in a glass is fine, but on a dress it becomes dirt. Shoes on the floor (Douglas' own example) are in order, but not on the table. Shaking hands or kissing can be civilized in one situation and outrageous and disgusting in another. Dirt, in other words, is disorder that causes psychological discomfort. Cleaning restores order.

This source extract explains that dirt, which is perceived depending on the context, causes psychological discomfort when out of place. Examples of out of place: when a drink is in a bottle, it is fine. However, when it is on say a piece of clothing, it is now out of place and therefore perceived as "dirt".

Source Extract: Staining of a Fabric Depend On Its Oleophobic and Hydrophobic Properties [17]

It becomes obvious that soil repellency and stain resistance, referring to the ability of a fabric to prevent liquid absorption and fine particles adherence to its surface, are directly linked to oleophobic and hydrophobic functionalities ([Textile Innovation Knowledge Platform, 2017](#)). In fact, it is mostly for marketing reasons that the scientific terms hydrophobic and oleophobic are translated for the consumer into buzzwords, such as soil repellent, stain resistant or self-cleaning ([Mahlting, 2014](#)).

This source extract explains what actually makes a fabric stain resistant, the oleophobic and hydrophobic functionalities. It also covers the common marketing terms used for consumers to better understand these terms.

Source Extract: What an R Factor of 2 Means in ISO 22196 [18]

Interpreting the results of ISO 22196 involves assessing the reduction in bacterial populations on treated surfaces. An effective antibacterial surface should show a significant decrease in bacteria, typically a logarithmic reduction. For ISO 22196, a log reduction of 2 or more is often considered effective, indicating a 99% reduction in bacteria.

This source extract explains that an R factor of 2 (log reduction of 2) for ISO 22196 is considered effective since it implies a 99% reduction in bacteria levels. This was particularly helpful to determine quantitatively what the numerical results of ISO 22196 mean.

Source Extract: Level of Efficacy of Antibacterial Activity [20]

Efficacy of antibacterial activity

From the testing result, the efficacy of antibacterial property of the test fabric can be considered as shown in [Table F.1](#).

Table F.1 — Efficacy of antibacterial property

Efficacy of antibacterial property	Antibacterial value: A
Low	$A < 2$
Significant	$2 \leq A < 3$
Strong	$A \geq 3$

For the efficacy of antibacterial property, taking into account variation in testing methods, operators, and testing laboratories, antibacterial value of 2,0 or greater is considered a level at which definite antibacterial effects can be confirmed, and is defined as a "significant effect."

Since different countries perceive antibacterial effects in different ways, this information has been provided in the annex not as a regulation but rather as reference information.

This source extract describes what different antibacterial values resulting from the test in ISO 20743 means. It was vital in determining the category of antibacterial the plastic in the design should fall under, as well as what antibacterial value achieves it.

Source Extract: AATCC 35 Sample Test Result [21]

Water Repellency

As shown in the table below, a 10.0 oz. cotton canvas treated with the suggested formulation mentioned previously can pass the rain test (AATCC 35) at the highest level typically used for apparel, 3 feet of water pressure for 5 minutes, even after 10 home launderings. Spray test ratings and Hydrostatic pressure test results are also shown in the table.

Sample - 100% cotton, 10.0 oz. canvas treated with fluorochemical water repellent	Unlaundered	After 10 HLTD ¹
Spray Test (AATCC 22) rating ²	100	80-85
Rain Test (AATCC 35) Blotter Weight Difference of less than 1.0 g = Pass Level: 3 feet for 5 minutes "storm"	0.2 g Pass	0.2 g Pass
Hydrostatic Pressure test (AATCC 127) – Suter ³	54.5 cm	N/A

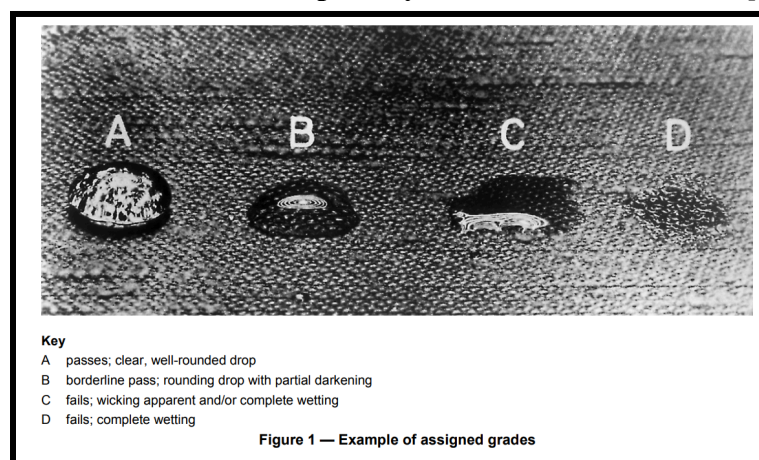
¹ "HLTD": Home laundering @ 105°F using standard AATCC detergent, and tumble dry for 30 minutes.

² Lowest acceptable spray rating is 70. Note that spray test ratings do not necessarily correspond to rain test results because the spray test only rates the surface of the fabric while the rain test measures the amount of water that passes through the fabric.

³ Testing performed by SGS U.S. Testing Company, Inc., Fairfield, NJ.

This source extract shows how a blot weight difference of less than 1.0 grams is considered the highest level result in apparel. This was vital to constructing objective 4.3 as the official AATCC test method 35 document only outlines the process for performing the test, omitting results and their implications.

Source Extract: Oil Repelency; Visual Demonstration [22]



This source extract provides a visual example for the levels of “oil resistance” of fabrics. This provides a simple way to evaluate fabrics put through the test method outlined in ISO 14419. From this, stakeholders were able to easily conclude what level of oleophobic they wished the design to have.

Appendicies

A.1 Chair Specifications

The specific chair that designs should work with is the “Very Side Chair,” designed by Haworth [25]. A CAD model can be found on their website [here](#) [25]. The dimensions of the chair are 22"D× 23.5"W×33.9"H [25], which can be seen in Figure A1.

A.2 Skeletal Muscle Definition

To clarify objective 2.3, “skeletal muscles” include every skeletal muscle that is entirely voluntarily controlled, such as the shoulder, hamstring, or abdominal muscles [26] [27]. This does not include cardiac or smooth muscle, as those are not considered skeletal muscles [26]. Cardiac and smooth muscle are essential for functions like pumping the heart or breathing—which must take place during sleep—so these are not included in the muscles that should be at rest (otherwise the user would have to be dead for the design to satisfy objective 2.3!). The diaphragm and extraocular muscles are excluded from muscles that should be at rest because these are active even during REM sleep [28].



Figure A1: The exact chair found in the EngSci common room that designs should work with, labelled with dimensions. [25]